Assignment 1: Derive the general equation of parallel projection onto a given view plane in the direction of a given projector $= a\hat{i} + b\hat{j} + c\hat{k}$. The view plane is defined by a reference point R(x_0,y_0,z_0) and the normal vector $N = n_1\hat{i} + n_2\hat{j} + n_3\hat{k}$. Give the solution as product of matrices, translation, x-axis rotation followed by y-axis rotation tp transform the view plane to the XY-plane. Then project the transformed point (x,y,z) on the XY-plane plane in the direction of v.

Hint: Obtain the full set of transformation matrix as you have done for arbitrary axis rotation.

To solve the above you may take help of the following:

Equation of parallel projection onto XY-plane in the direction of projection $v = a\hat{i} + b\hat{j} + c\hat{k}$.

Solution: Let $A(x, y, z)$ be any point and $B(x_p, y_p, z_p)$ is the parallel projection of A on XY plane. The vector $\vec{AB}$ is defined as

$$\vec{AB} = (x_p - x)\hat{i} + (y_p - y)\hat{j} + (z_p - z)\hat{k}$$

Since $\vec{AB}$ is parallel to V, this means $\vec{AB} = v$ where t is a constant.

$$(x_p - x)\hat{i} + (y_p - y)\hat{j} + (z_p - z)\hat{k} = t(a\hat{i} + b\hat{j} + c\hat{k})$$

i.e.

$$x_p - x = at$$
$$y_p - y = bt$$
$$z_p - z = ct$$

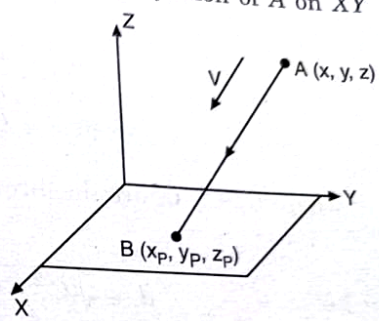

Fig. 11.12.

Since point $B(x_p\ y_p\ z_p)$ falls on xy plane, So $z_p = 0$

$$\therefore \quad z_p - z = ct \Rightarrow 0 - z = ct \quad \text{or} \quad t = -\frac{z}{c}$$

Put the value of t, we get

$$x_p - x = -a \cdot \frac{z}{c} \Rightarrow x_p = x - \frac{az}{c}$$

$$y_p - y = -b \cdot \frac{z}{c} \Rightarrow y_p = y - \frac{bz}{c}$$

and $$z_p = 0.$$

So, the transformation matrix in 4 × 4 form is

$$\begin{bmatrix} x_p \\ y_p \\ z_p \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & -a/c & 0 \\ 0 & 1 & -b/c & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$